MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Uses of an Oscilloscope

4.6

STUDENT NAME

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PHYSICS WITH MOIZ DAWOODI

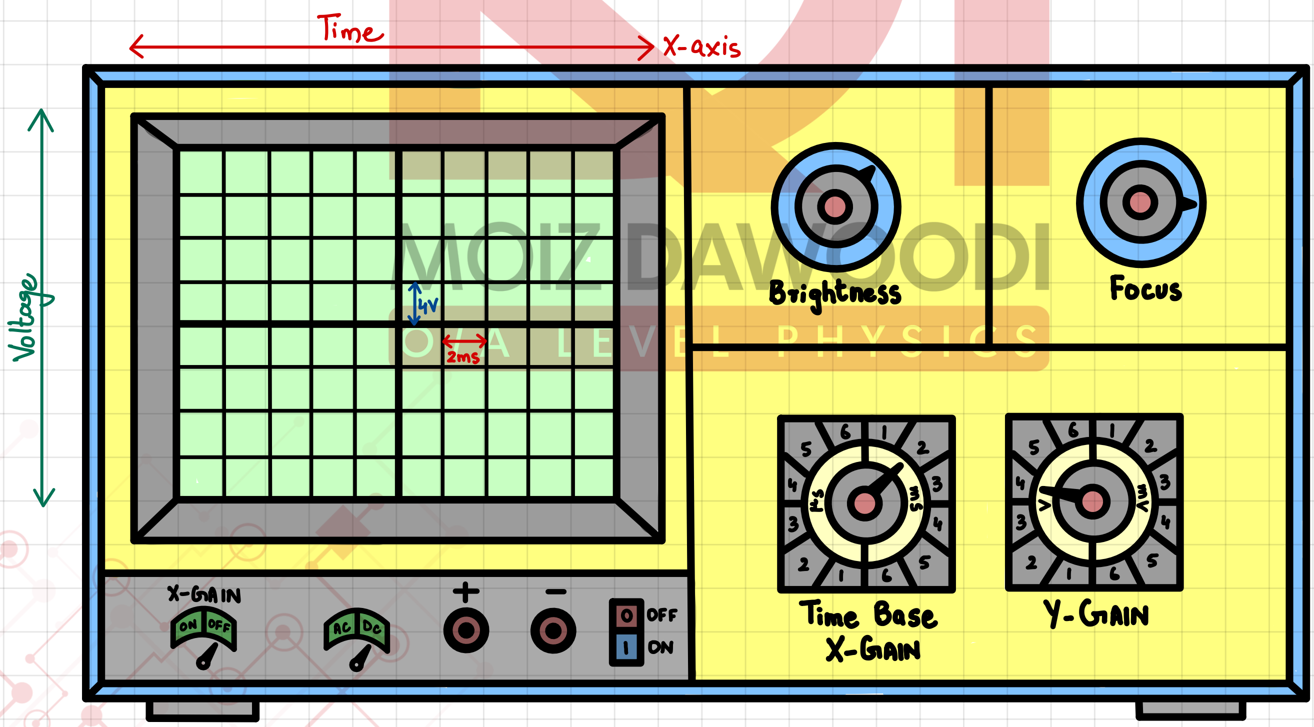
Uses of an Oscilloscope

SYLLABUS OUTLINE:

4.6	Uses of an oscilloscope
1	Describe the use of an oscilloscope to display waveforms (the structure of an oscilloscope is **not** required)
2	Describe how to measure p.d. and short intervals of time with an oscilloscope using the Y-gain and timebase

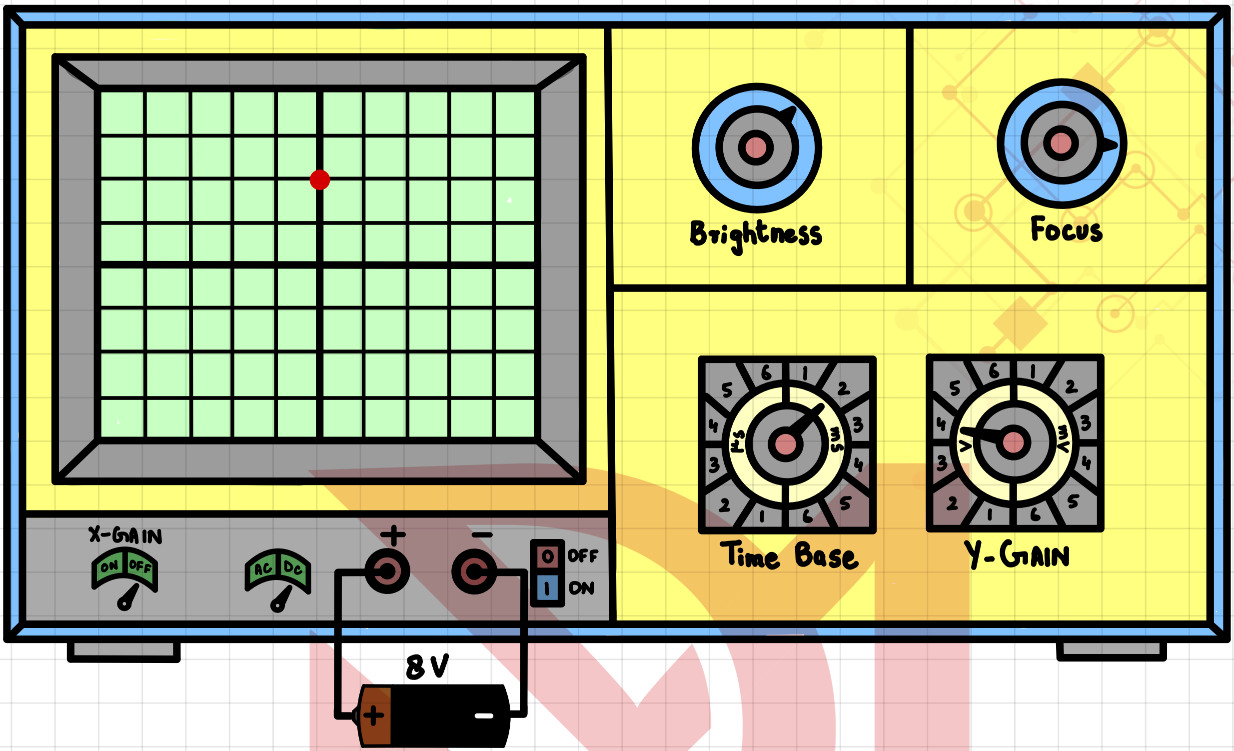
Cathode Ray Oscilloscope (CRO):

Cathode ray Oscilloscope is an electronic device which is used to visualize waveforms, measures voltage, measures time period etc.

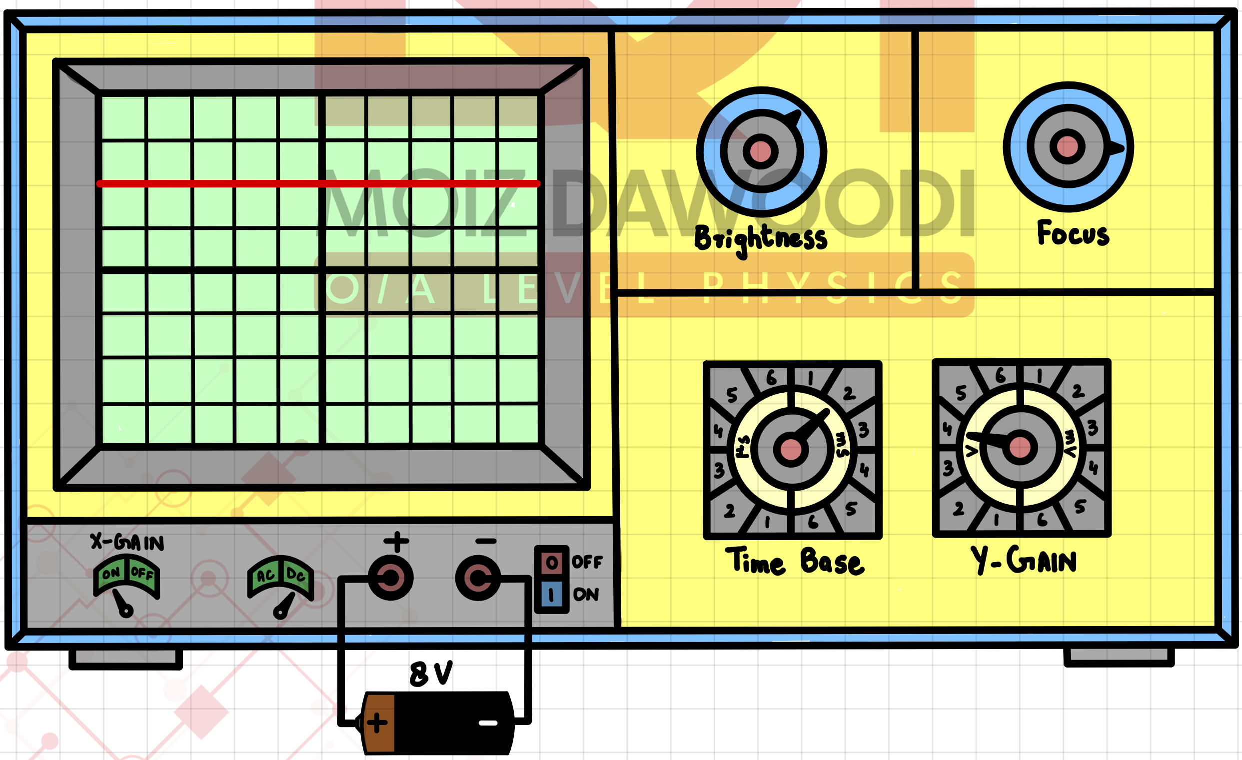


TRACES:

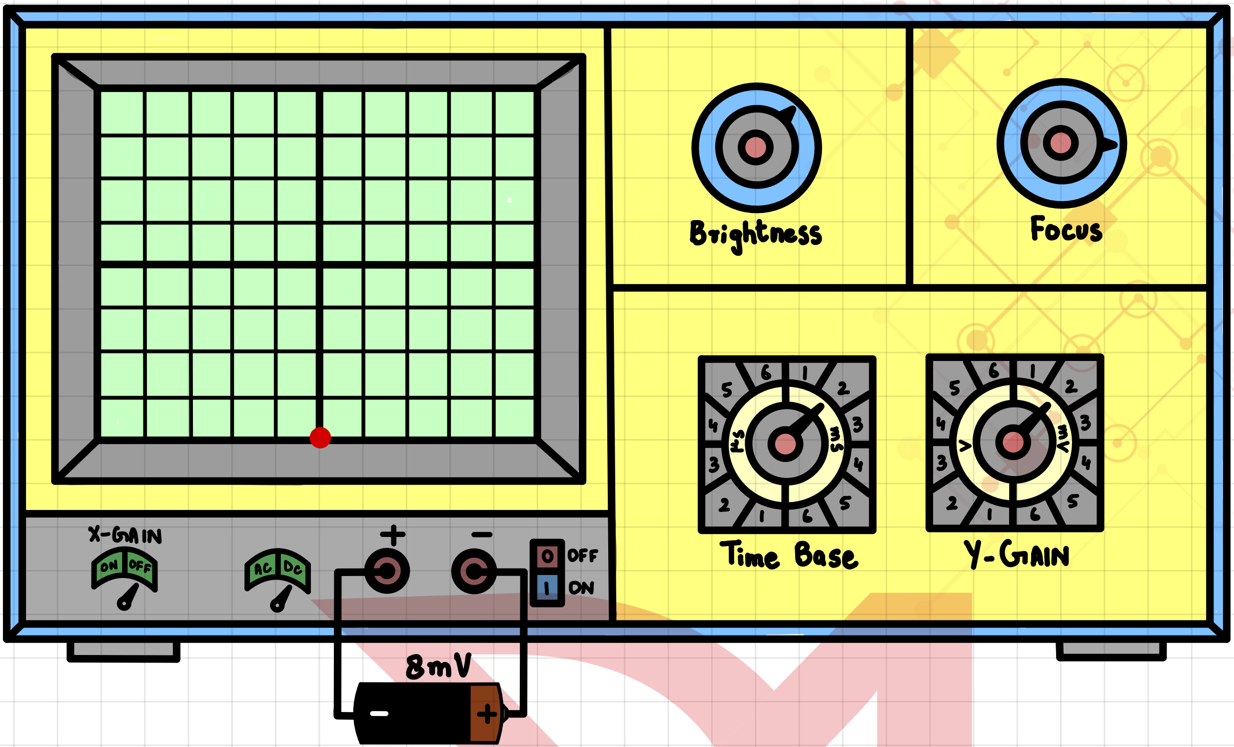
- Trace of an 8V D.C. source connected with correct polarity with X-gain OFF.



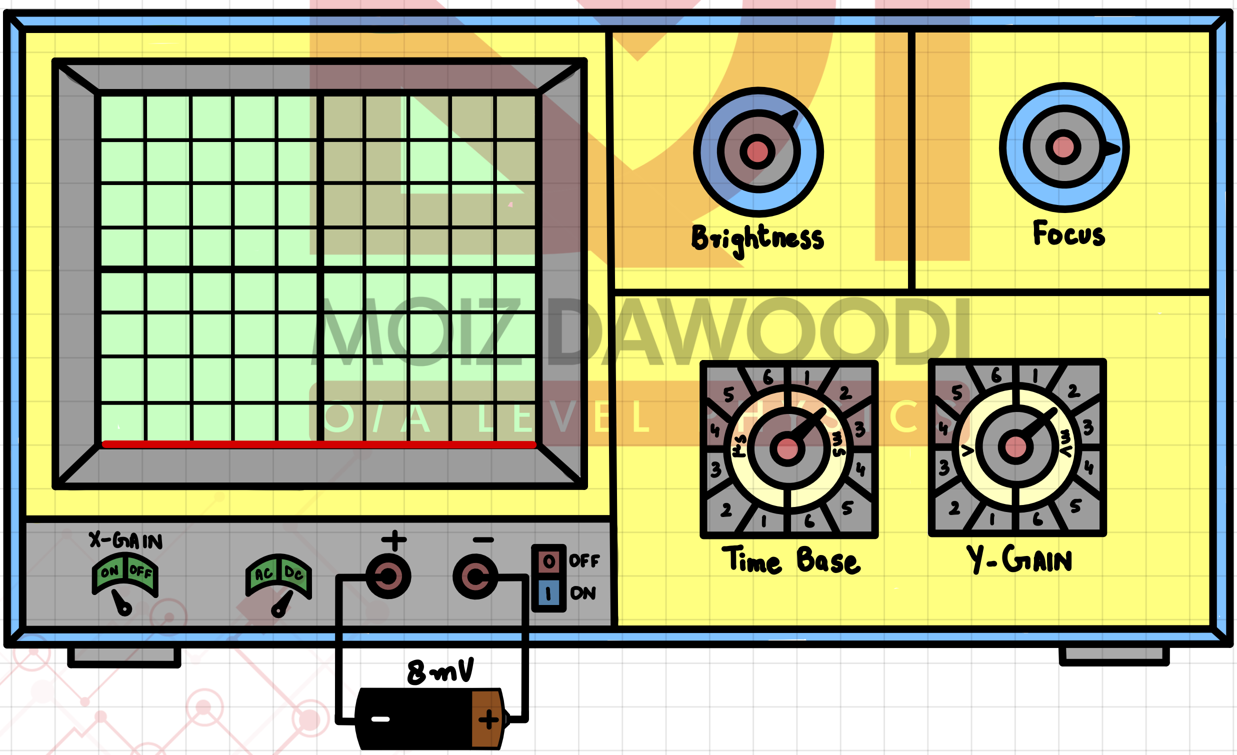
- Trace of an 8V D.C. source connected with correct polarity with X-gain ON.

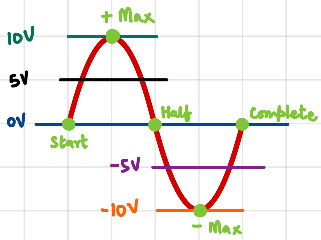


- Trace of an 8mV D.C. source connected with opposite polarity with X-gain OFF.



- Trace of an 8mV D.C. source connected with opposite polarity with X-gain ON.





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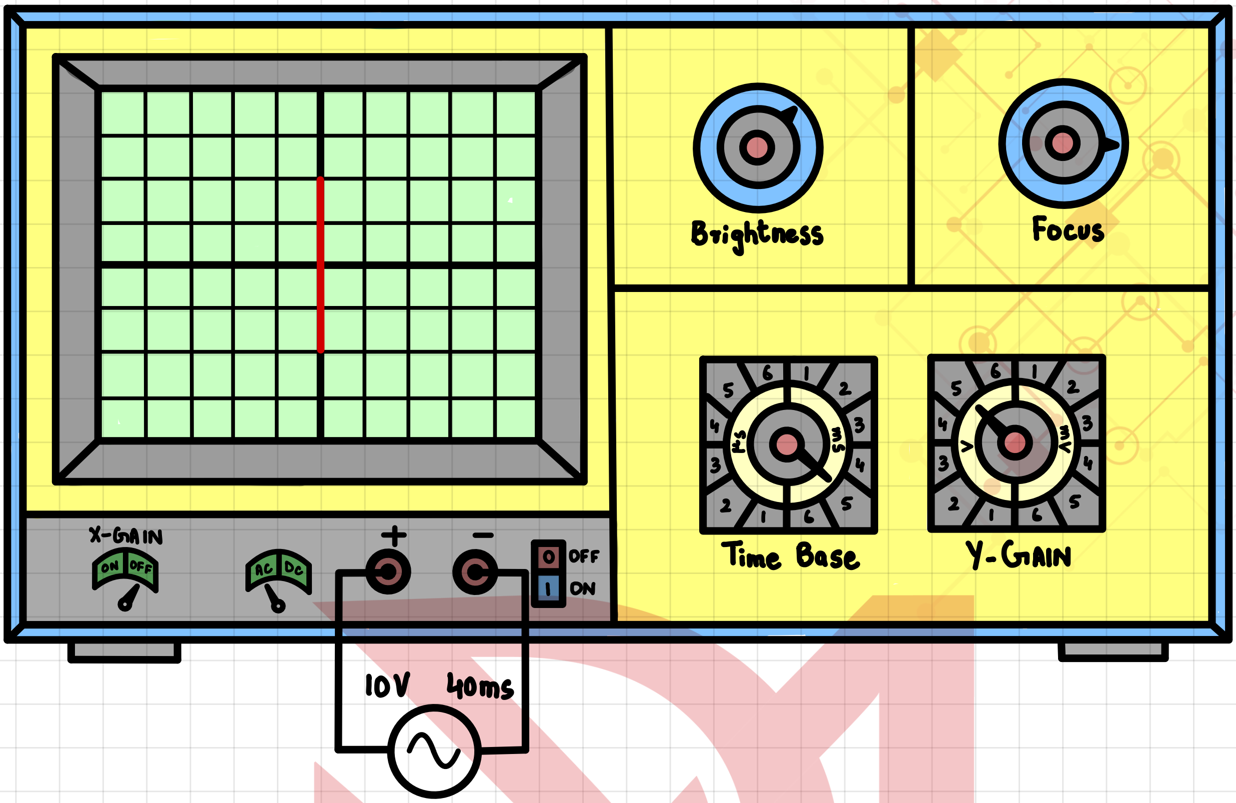
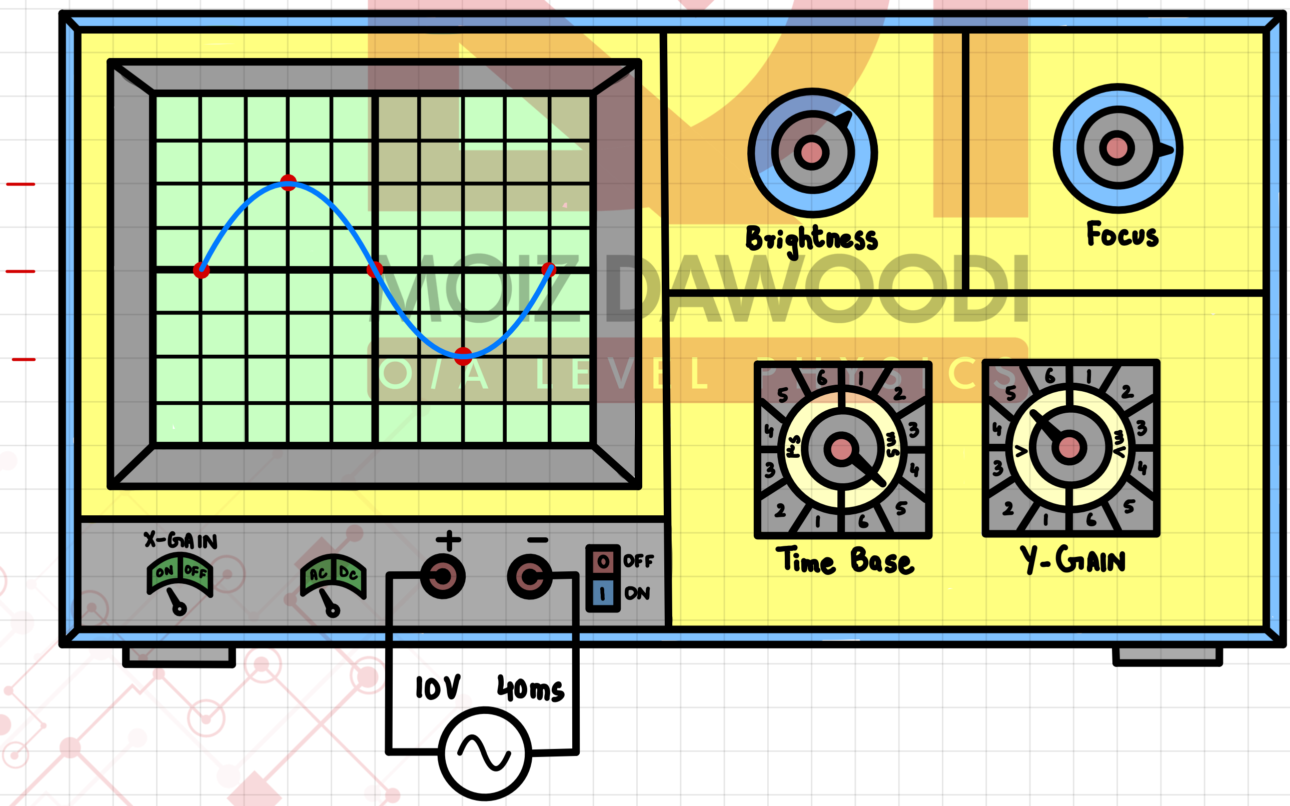
- Trace of an 10V, 40ms A.C. source connected with X-gain OFF.



- Trace of an 10V, 40ms A.C. source connected with X-gain ON.



Match the following

SCAN ME

MATCH THE FOLLOWING TERMINOLOGIES WITH CORRECT INFORMATION.

CRO is used	Y-axis on CRO screen
Voltage is along	X-axis on CRO screen.
Time is along	to visualize waveform.
Y-gain of 5V/div	No trace is seen on x-axis.
X-gain of 2ms/div	donot move along y-axis.
Trace of DC	means 2ms represented by 1 box on x-axis
If time base is OFF	means 5V represented by 1 box on y-axis

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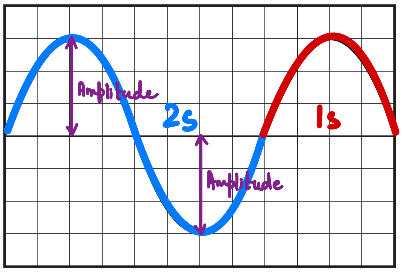
1 An alternating voltage of frequency 0.5 Hz is applied to the Y-plates of a cathode-ray oscilloscope (c.r.o.).

$T = \frac{1}{f} = \frac{1}{0.5} = 2\,s$

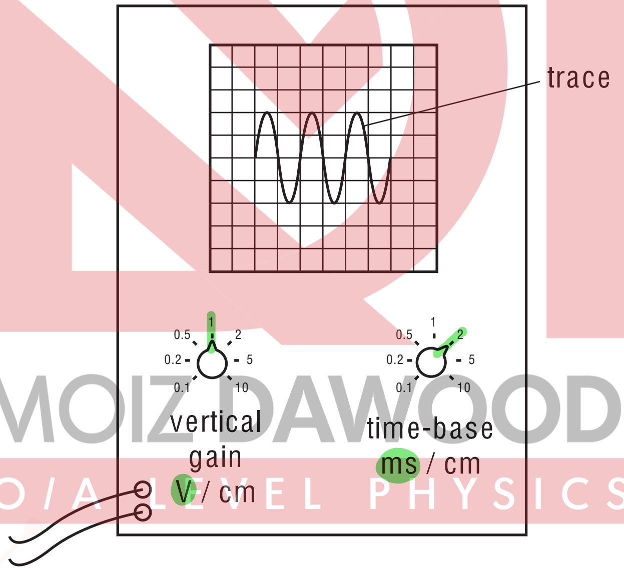
The diagram shows the screen of the c.r.o.



What is the time taken for the spot to cross the screen?

A 3 s **B** 6 s **C** 15 s **D** 30 s [MJ2011/P11/Q37]

2 The trace of a waveform is seen on the screen of a cathode-ray oscilloscope.



Which statement about the controls is correct?

A The amplitude of the trace is changed by adjusting the time-base. ✗

B The amplitude of the trace is changed by adjusting the vertical gain.

C The whole trace is moved to the right by adjusting the time-base. ✗ x-shift

D The whole trace is moved upwards by adjusting the vertical gain. ✗ y-shift [ON2012/P12/Q37]

P 1 Mark Scheme

1	A
2	B

1 A microphone is connected to a cathode-ray oscilloscope (c.r.o.). A note produced by a musical instrument causes a trace on the screen. The trace is stored electronically in the c.r.o.

Fig. 1.1 shows the stored trace displayed on the screen.

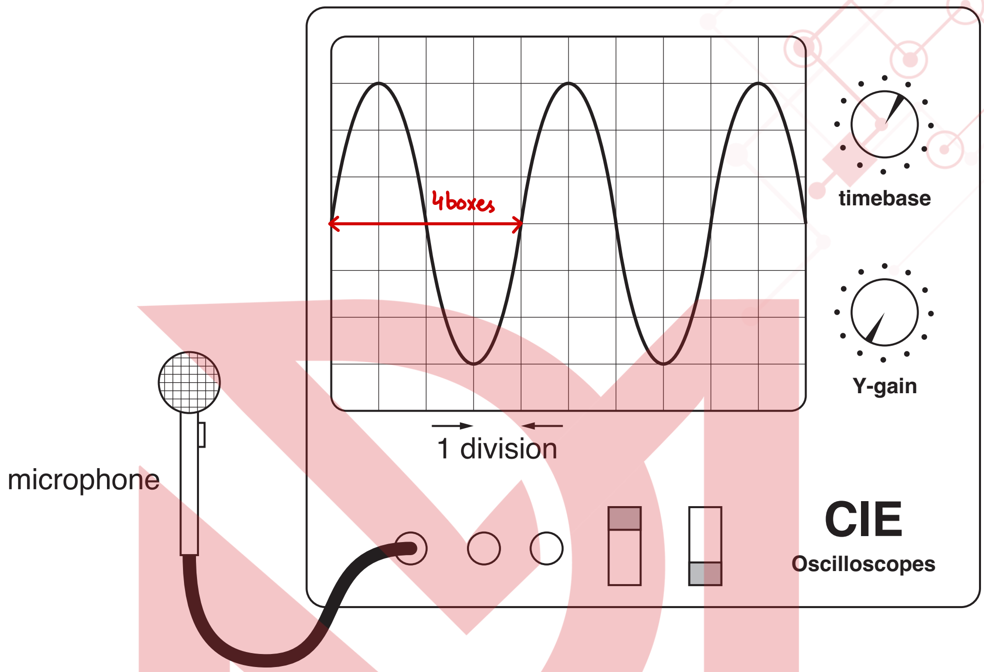


Fig.1.1

The timebase setting on the oscilloscope is 0.20 ms / division.

(a) Determine the frequency of this note.

$T = 4 \times 0.2\,ms = 0.8\,ms$ → 10^{-3}

$f = \frac{1}{T} = \frac{1}{0.8 \times 10^{-3}} = 1250\ Hz$

frequency = 1250 Hz ....[3]

[ON2010/P22/Q5]

2 **(c)** The output terminals of the a.c. generator are connected to a cathode-ray oscilloscope (c.r.o.). Fig. 2.2 shows the trace on the screen of the c.r.o.

Y-gain: 5V/div

X-gain: 2ms/div

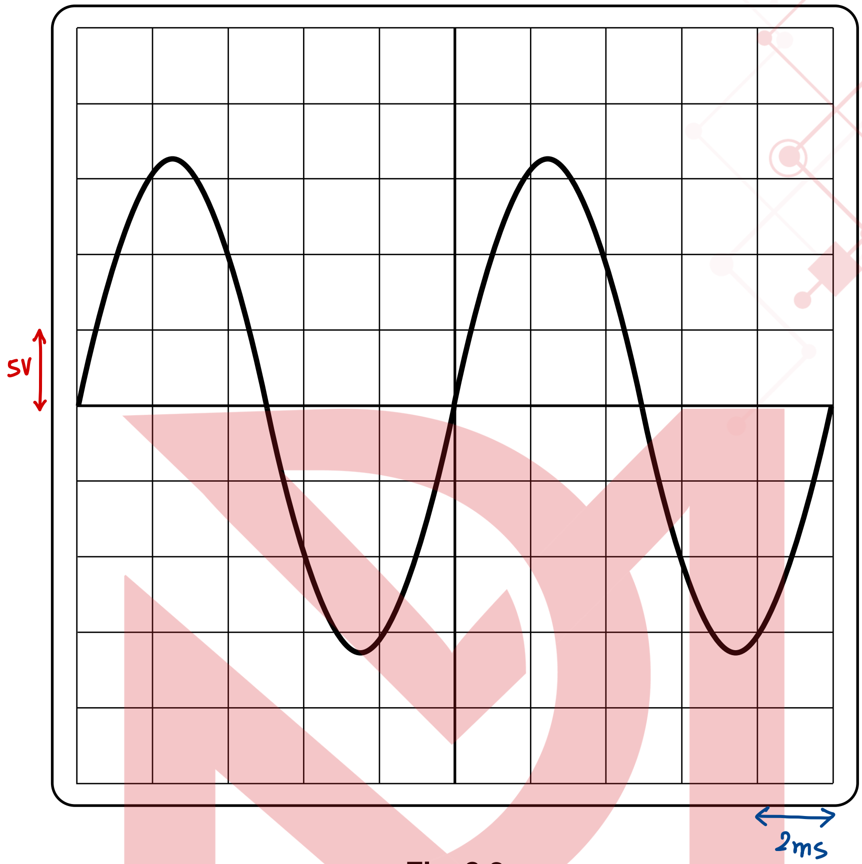


Fig. 2.2

Describe how the trace and a setting on the c.r.o. are used to find the time for one revolution of the coil of the a.c. generator. You may draw on Fig. 1.2 if you wish.

Observe the time base setting of CRO (e.g: 2ms/div) & multiply it by 5.

....[2]

[ON2014/P21/Q8]

3 **EITHER**

Fig. **3**.1 shows a cathode-ray oscilloscope (c.r.o.). An alternating voltage supply is applied to the input and the trace produced is shown.

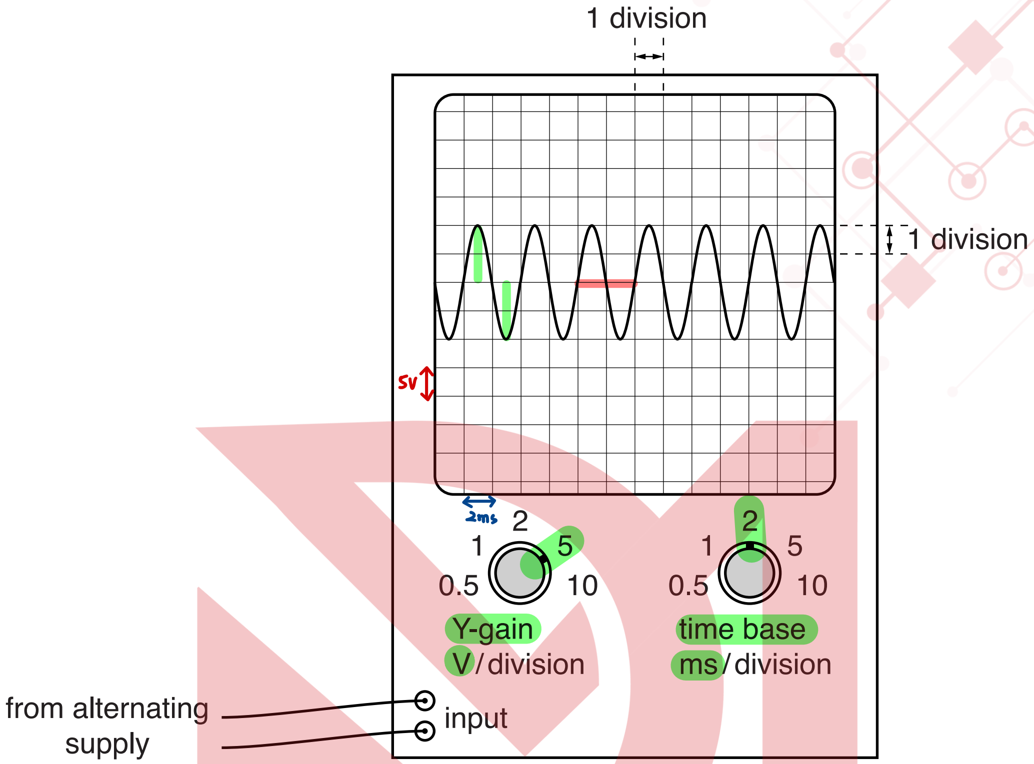


Fig. 3.1

(a) Use Fig. **3**.1 to determine the maximum voltage of the supply.

5V x 2

voltage = 10V [2]

(b) Describe how the trace on the screen and the controls on the c.r.o. are used to measure the time between any two points on the trace.

Observe time base setting i.e. 2ms/div & multiply it by 2 as wave takes 2 divisions along x-axis to complete.

[2]

[MJ2016/P21/Q9]

4 **(b)** A microphone is connected to a c.r.o. to display a sound wave.

Fig. 4.2 shows the trace on the c.r.o.

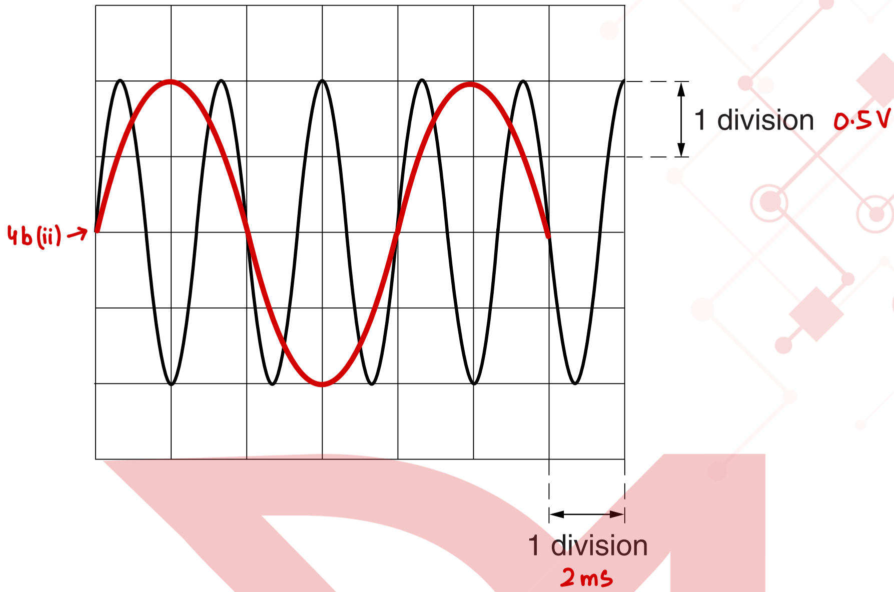


Fig. 4.2

The settings on the c.r.o. are: Y-gain 0.5 V / division; time base 2.0 ms / division.

(i) Determine

1. the maximum voltage,

0.5 V x 2

voltage = 1.0 V [1]

2. the time for one oscillation,

3 waves → 8ms (4 boxes)

1 wave → T

$T = \frac{8ms}{3}$

m = 10^{-3}

ms = 10^{-3} s

time = 2.66 ms or 2.66×10^{-3} s [2]

3. the frequency of the sound wave.

$f = \frac{1}{T} = \frac{1}{2.66 \times 10^{-3}} = 375.93$

frequency = 376 Hz [2]

(ii) The settings of the c.r.o. remain the same. On Fig. 4.2, sketch the trace of a sound wave with a smaller amplitude and a lower frequency. [2]

[MJ2016/P22/Q10]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2010	5054	22

5 (a) 0.80 **or** 0.0008 **or** 4 × 0.20 **or** 4 × 0.0002 **or** 4 divisions C1
(f =) 1/T **or** 1.2/1.25/1.3 (Hz) C1
1200/1250/1300 Hz A1

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2014	5054	22

(c) (half) distance across screen **or** count divisions of / measure wavelength **or** the wavelength corresponds to one rotation B1
half distance multiplied by time base setting B1

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	21

9 E (a) 2 squares C1
10 V A1

(b) measure / find horizontal distance / number of divisions (between points) C1
distance × no (m)s / division A1

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	22

(b) (i) 1 1(.0) V B1

2 one wave 1.3–1.4 squares **or** 3 waves in 4 squares C1
2.6–2.8 ms A1

3 (f =) 1 / T numerical or algebraic C1
345–400 Hz A1

(ii) smaller amplitude shown B1
larger period shown B1